UNIVERSITY OF BIRMINGHAM | GRAVITATIONAL WAVE ASTRONOMY

OBSERVATIONAL COSMOLOGY

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RECAP

- Friedmann equation

$$\left(\frac{\dot{a}(t)}{a(t)}\right)^2 = H(t)^2 = \frac{8\pi G}{3}\rho(t) - \frac{\kappa c^2}{R_0^2 a^2(t)}$$

- Fluid equation

$$\dot{\rho}(t) + 3\frac{\dot{a}(t)}{a(t)}\left(\rho(t) + \frac{p}{c^2}\right) = 0$$

- Equation of state (EOS)

$$p = w\rho c^2$$

- Acceleration equation

$$\frac{\ddot{a}(t)}{a(t)} = -\frac{4\pi G}{3}\left(\rho(t) + \frac{3p}{c^2}\right)$$

THE CRITICAL DENSITY

- The critical (or closure) density $\rho_c(t)$ is <u>defined such that the universe is flat</u>, i.e. $\kappa = 0$:

$$\left(\frac{\dot{a}(t)}{a(t)}\right)^2 = H(t)^2 = \frac{8\pi G}{3}\rho(t)$$

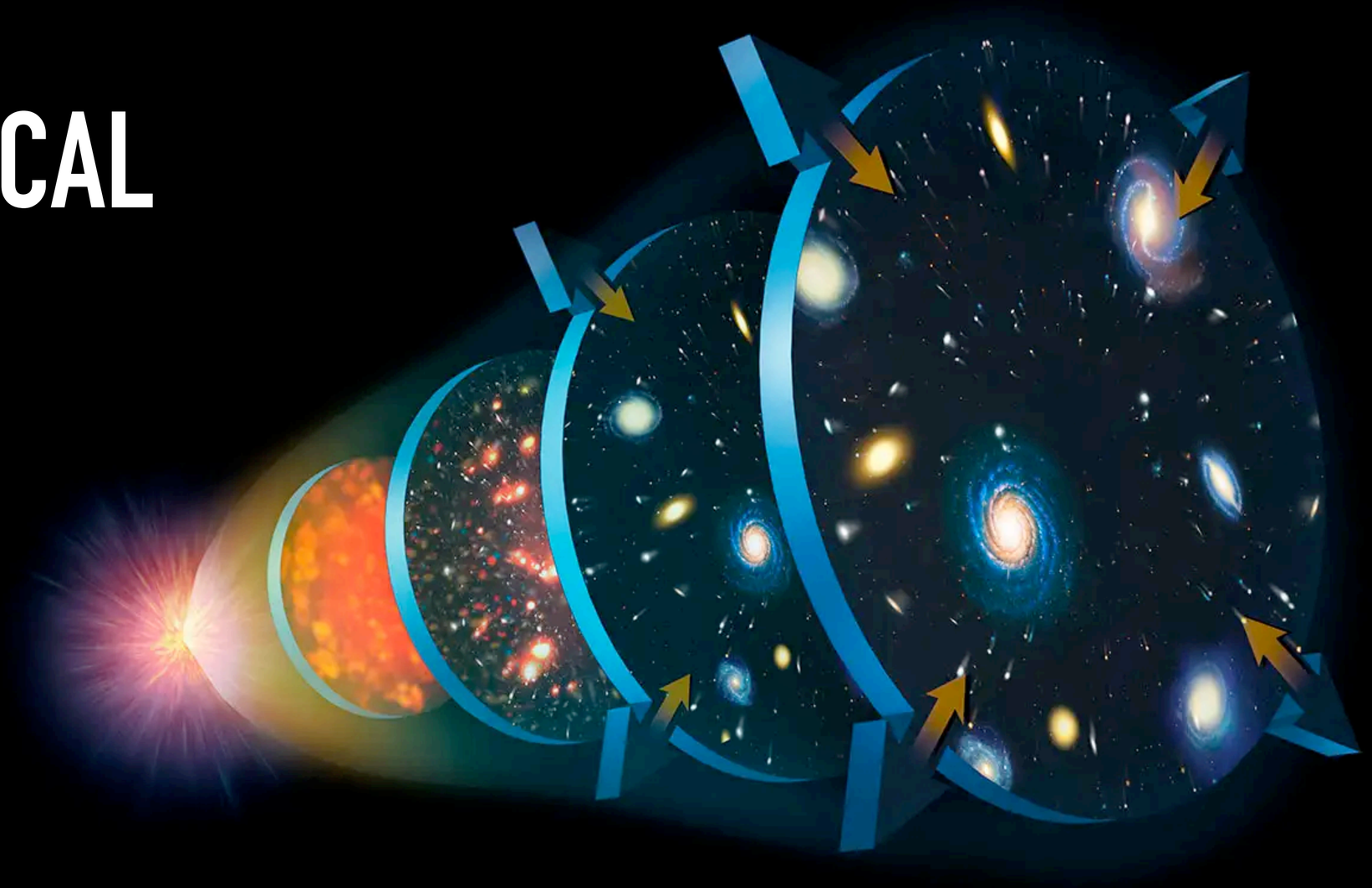
- Thus, the **critical density** $\rho_c(t)$ is given by:

$$\boxed{\rho_c(t) = \frac{3H(t)^2}{8\pi G}}$$

- Assuming today's value for the Hubble parameter $H_0 = 68\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (from the CMB measurement), the critical density at the current epoch is:

$$\rho_c(t_0) = \rho_{c,0} = 8.7 \times 10^{-27}\,\mathrm{kg\,m^{-3}} \approx 10^{-28}\,\mathrm{kg\,m^{-3}}$$

COSMOLOGICAL MODELS



MODEL UNIVERSES: SOLVING THE EVOLUTION EQUATIONS

?? What is the time evolution (fate) of some (model) universe ??

- A <u>model universe</u> is a solution of the Friedmann equation under certain assumptions about its contents (i.e. $\rho(t)$) and its spatial geometry (i.e. κ).
  - The Friedmann and fluid equations are coupled, i.e., we need to solve the fluid equation in order to solve the Friedmann equation.

- Let's start with solving the fluid equation:
  - Energy/matter components to consider are:
    - Non-relativistic matter, e.g. baryons, cold dark matter: $w = 0$
    - Relativistic matter, e.g. photons, neutrinos: $w = 1/3$
    - Dark energy with $w = -1$ (cosmological constant)
    - Other exotic forms of matter: some other value for w

SOLVING THE FLUID EQUATION

- Suppose the universe contains N different matter/energy components ("fluids") $\rho_i(t)$
- The **total energy/matter density** and total pressure are

$$\rho(t) = \sum_{i=1}^{N} \rho_i(t), \quad p = \sum_{i=1}^{N} w_i \rho_i(t) c^2$$

- **Note:** The fluid equation must hold for each component individually if there is no interaction between "fluids"
- Rearranging the fluid equation yields:

$$\dot{\rho}_i(t) + 3\frac{\dot{a}(t)}{a(t)}(1 + w_i)\rho_i(t) = 0$$

- The general solution to this first order ODE is:

$$\rho_i(a(t)) = \rho_i(t_0) a(t)^{-3(1+w_i)} = \rho_{i,0} a(t)^{-3(1+w_i)}$$

- For matter: $\rho_m(t) = \rho_{m,0} a(t)^{-3}$ For radiation: $\rho_r(t) = \rho_{r,0} a(t)^{-4}$

SINGLE-COMPONENT UNIVERSES

- We then use the solutions of the fluid equation to solve the Friedmann equation

- Flat, radiation-only solution:

$$\rho_r(t) = \frac{\rho_{r,0}}{a(t)^4}$$

$$a(t) = \left(\frac{t}{t_0}\right)^{1/2}$$

$$\rho_r(t) = \frac{\rho_{r,0}}{a(t)^4} = \frac{\rho_{r,0} t_0^2}{t^2}$$

$$H(t) = \frac{1}{2t}$$

- Flat, matter-only solution:

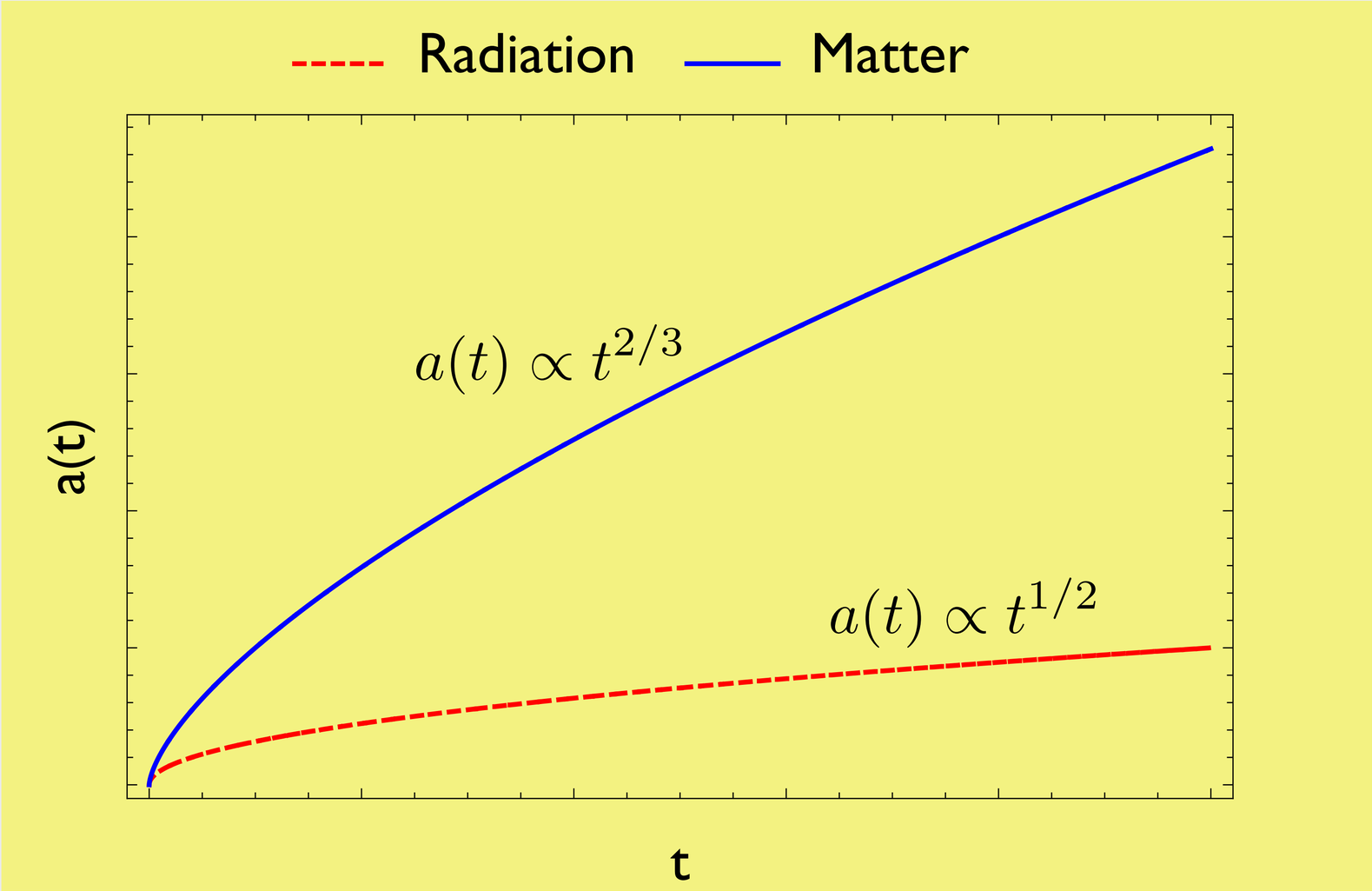
$$\rho_m(t) = \frac{\rho_{m,0}}{a(t)^3}$$

$$a(t) = \left(\frac{t}{t_0}\right)^{2/3}$$

$$\rho_m(t) = \frac{\rho_{m,0}}{a(t)^3} = \frac{\rho_{m,0} t_0^2}{t^2}$$

$$H(t) = \frac{2}{3t}$$

SINGLE-COMPONENT UNIVERSES



A MATTER-RADIATION MIXTURE UNIVERSE

- **Flat mixture model**: $\rho(t) = \rho_m(t) + \rho_r(t)$ and $\kappa = 0$
- Friedmann equation:

$$H(t)^2 = \frac{8\pi G}{3}\rho(t) = \frac{8\pi G}{3}\left(\frac{\rho_{m,0}}{a^3(t)} + \frac{\rho_{r,0}}{a^4(t)}\right)$$

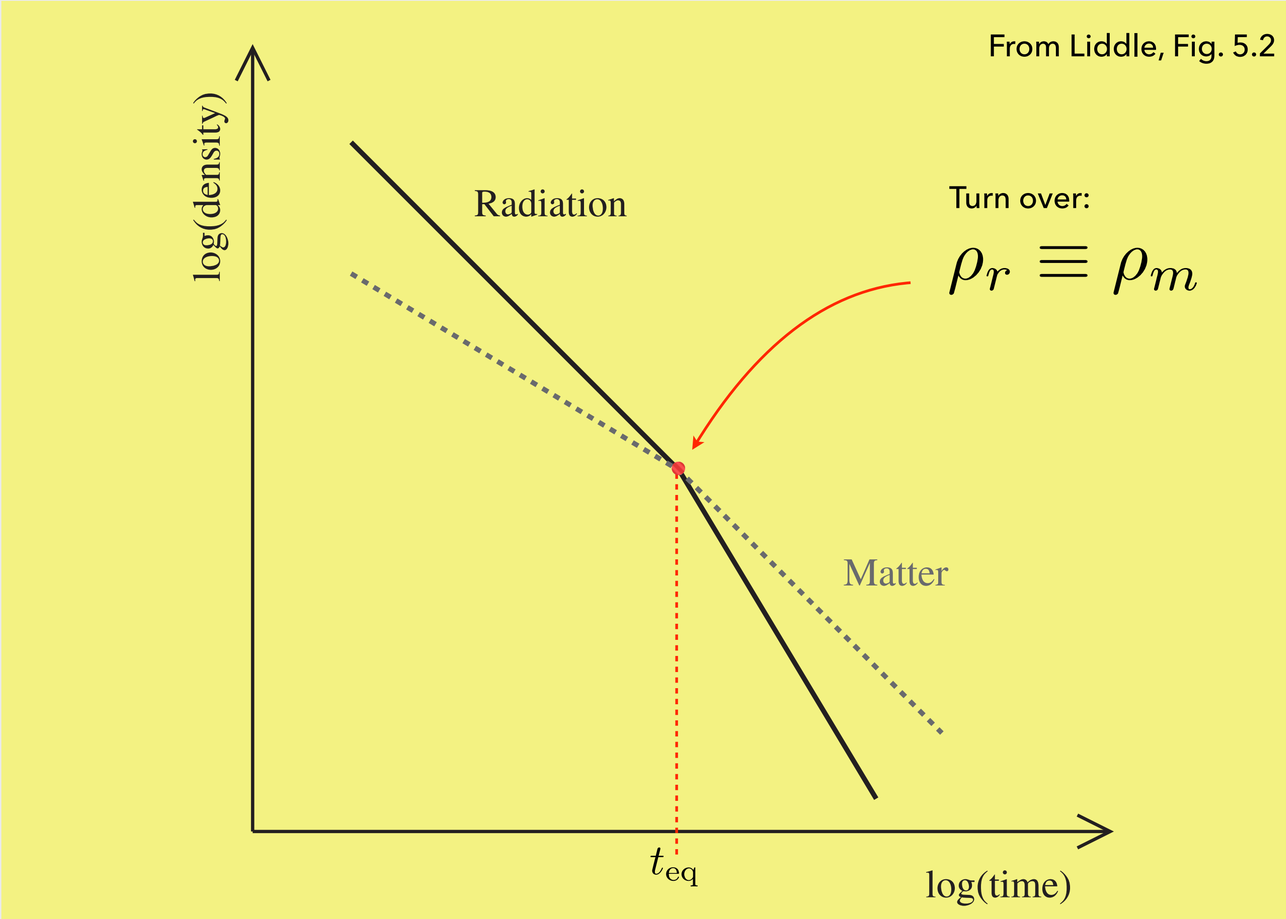
- Rather than solving the full Friedmann equation, we look at it in different "regimes" (limits). This is possible as each component depends differently on the scale factor.
- In the radiation dominated regime, we find:

$$a(t) = (t/t_0)^{1/2} \quad \Rightarrow \rho_m(t) = \rho_{m,0}(t_0/t)^{3/2}$$

- In the matter-dominated regime, we find:

$$a(t) = (t/t_0)^{2/3} \quad \Rightarrow \rho_r(t) = \rho_{r,0}(t_0/t)^{8/3}$$

A MATTER-RADIATION MIXTURE UNIVERSE



"equipartition" or radiation-matter-equality

Figure 5.2 A schematic illustration of the evolution of a universe containing radiation and matter. Once matter comes to dominate the expansion rate speeds up, so the densities fall more quickly with time.